NLP end-to-end and embeddings

WikiText-103, SGNS, gensim, GloVe and AG News

1. Research and preprocessing

PyTorch `Embedding` maps IDs to rows through `num_embeddings` and `embedding_dim`. `padding_idx` reserves an untrained zero row; `sparse=True` saves gradient storage with a compatible optimizer. `from_pretrained` imports weights and `freeze` controls adaptation. `EmbeddingBag` uses flattened IDs and offsets for `mean`, `sum` or `max`; the classifier uses mean. CBOW predicts a center from context; skip-gram predicts context from a center. SGNS retains separate input and output tables, exporting input vectors. Word2Vec is predictive; GloVe fits weighted global log co-occurrences. Gensim uses `sg=1`, `negative`, `sample`, `alpha`, `epochs` and fixed windows; `KeyedVectors` supports persistence and cosine lookup.

WikiText train contains **29,444 articles, 1,801,350 lines and 85,929,215 normalized body tokens**. Validation has 182,322 tokens and test 204,879; neither trains embeddings. Article counts use top-level titles. Headings and standalone punctuation are excluded. Lowercase words retain apostrophes and numeric strings, repair WikiText markers, and split hyphens. This matches uncased GloVe without inventing a numeric token. Sentence splitting remains approximate for abbreviations.

The notebook compares 11 sentences with NLTK Treebank: `can't` stays one word but Treebank yields `ca` and `n't`; compounds split only in the custom tokenizer; abbreviations lose dots; numeric separators stay internal. All AG News stages reuse this exact tokenizer.

min_count	Vocabulary	Unknown tokens %
5	80,087	1.295
10	54,175	2.145
20	36,728	3.337

The vocabulary is fixed from the 20M subset. Top 10/1,000/30,000 words cover 24.44% / 64.62% / 95.51% of train tokens. The top-30,000 log-log slope is -1.274: Zipf-like, not an exact slope of -1. With $t = 10^{-4}$, observed discard rates for the/of/and are 96.26% / 94.13% / 93.80%. This reduces function-word dominance. Fixed-window pairs fall from 337,969,464 to 180,580,964 per epoch in the 40M subset; contexts never cross sentence boundaries.

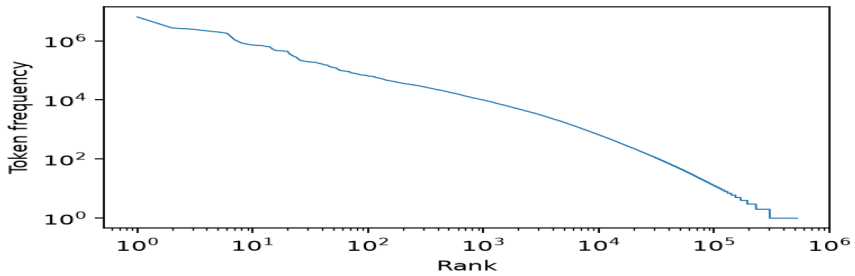


Figure 1: Observed frequency versus rank.

2. SGNS implementation and controlled iterations

Two sparse embedding tables minimize positive and negative logistic losses. Noise probability is proportional to frequency raised to 0.75. Batched summed gradients use SGD, linearly decaying learning rate 0.025 to 0.0001 over three epochs, `min_count=5`, subsampling 10^{-4} and batch size 2,048. Gensim trains identical encoded sentences, vocabulary and equivalent settings. It differs through asynchronous online updates and skipped negative collisions. Loss values across implementations or different negative counts are not directly comparable.

Run	M tok	Dim	Win	Neg	Best ep	Loss	Add	WS	Sec	GPU MB
ref	40.00	100	5	5	3	2.206	0.433	0.681	581	88
dim50	40.00	50	5	5	3	2.233	0.333	0.637	518	54
dim300	40.00	300	5	5	3	2.173	0.469	0.680	450	232
20M	20.27	100	5	5	3	2.213	0.310	0.634	191	88
30M	30.00	100	5	5	3	2.208	0.383	0.667	269	88
win2	40.00	100	2	5	3	2.125	0.397	0.633	154	88
neg10	40.00	100	5	10	3	2.763	0.447	0.668	342	96

Metrics in the table refer to each run's best epoch; time sums all three epochs, and memory is peak PyTorch allocation, not total GPU process memory. The notebook retains every epoch's semantic/syntactic/total accuracy, WordSim-353 Spearman, loss, timings and five neighbors of king, france, computer, good, january and run.

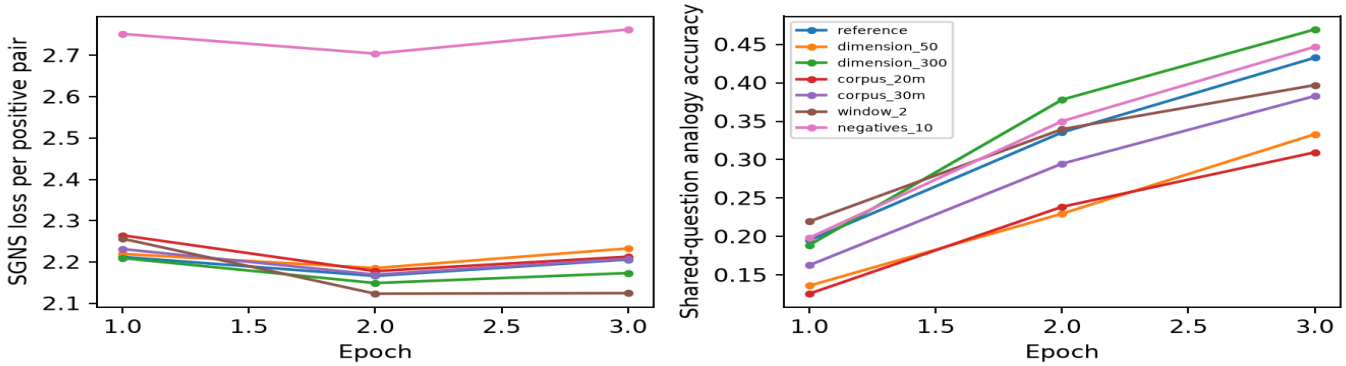


Figure 2: Loss and fixed-question analogy accuracy for seven runs.

The selected model is **dimension_300, epoch 3**. Selection uses only shared-question 3CosAdd accuracy then WordSim-353. SimLex and AG News test are held out. Best analogy accuracy for dimensions 50/100/300 is 0.333/0.433/ 0.469. For 20M/30M/40M tokens it is 0.310/0.383/0.433. Window two yields 0.397; ten negatives yield 0.447. These are descriptive single-seed comparisons, not statistical significance tests. In the reference run, loss changes from 2.166 at epoch two to 2.206 at epoch three, while analogy accuracy changes from 0.335 to 0.433. A lower negative-sampling loss is not an unconditional quality guarantee: the objective and benchmarks measure different properties. Cross-negative-count losses also change their number of terms.

3. Vector arithmetic, coverage and parallelism

Custom 3CosAdd uses unit source vectors and normalized $b - a + c$, excluding query words. Its top five match gensim numerically. The notebook evaluates 18 original analogies across six types, retaining all similarities and correct-answer ranks. For man → woman, king → queen:

Model	Top five with cosine	Queen rank	Cosine	First, no exclusion
sgns	queen 0.58; consort 0.50; coronation 0.49; anointed 0.48; princess 0.48	1	0.578	king
gensim	queen 0.55; anointed 0.50; electress 0.50; consort 0.50; lfgifu 0.49	1	0.555	king
glove	queen 0.77; monarch 0.68; throne 0.68; daughter 0.66; princess 0.65	1	0.770	king

Across all original analogies: sgns: 17/18 top-one correct; gensim: 14/18 top-one correct; glove: 17/18 top-one correct.

The benchmark uses the same 30,000 most frequent shared words. Coverage is 12,322/19,544 questions. S/G/V denote SGNS/gensim/GloVe. Every category reports evaluated/total:

Category	Coverage	S Add	G Add	V Add	S Mul	G Mul	V Mul
capital-common	420/506	0.560	0.576	0.933	0.571	0.586	0.917
capital-world	1166/4524	0.400	0.382	0.895	0.391	0.381	0.866
currency	108/866	0.028	0.009	0.046	0.037	0.028	0.037
city-in-state	1811/2467	0.308	0.292	0.298	0.301	0.288	0.261
family	342/506	0.670	0.696	0.883	0.635	0.681	0.874
g1-adjective-to-adverb	812/992	0.151	0.148	0.278	0.132	0.134	0.280
g2-opposite	306/812	0.190	0.170	0.281	0.147	0.131	0.245
g3-comparative	1332/1332	0.633	0.622	0.803	0.634	0.617	0.780
g4-superlative	702/1122	0.330	0.318	0.613	0.358	0.363	0.618
g5-present-participle	870/1056	0.439	0.439	0.731	0.420	0.429	0.721
g6-nationality-adjective	1299/1599	0.838	0.856	0.973	0.848	0.868	0.958
g7-past-tense	1406/1560	0.591	0.605	0.576	0.579	0.589	0.550
g8-plural	992/1332	0.458	0.513	0.794	0.441	0.492	0.754
g9-plural-verbs	756/870	0.373	0.352	0.585	0.386	0.368	0.608

The missing section 5.3 is interpreted as cosine parallelism of raw difference vectors from independent pairs of the same relation. Shared-word comparisons are excluded. Mean cosine and comparison counts are sgns: 0.305, 90 comparisons; gensim: 0.297, 90 comparisons; glove: 0.591, 90 comparisons.

The strongest SGNS category is gram6-nationality-adjective at 0.838; the weakest is currency at 0.028. Query words can dominate without exclusion, and difference vectors are not perfectly parallel. Consequently the familiar king−man+woman expression is approximate retrieval, not equality. A 500-word thematic t-SNE is in the notebook: technology and countries form recognizable groups, while verbs and animals overlap more. Neighborhood-based theme selection itself favors visible grouping.

4. End-to-end AG News classification

The official 120,000 training documents split into 108,000 train and 12,000 stratified validation documents. TF-IDF fits train only. Mean EmbeddingBag feeds a 128-unit ReLU/dropout MLP; missing words map to ignored padding, with a zero representation for all-unknown documents. Two learning rates and validation macro F1 select epochs and frozen/fine-tuned variants. TF-IDF tunes two values of C using warm-start logistic regression; its loss curve uses optimizer blocks. The selected SGNS/gensim dimension differs from GloVe's 100D; classification differences therefore include a dimension confound. Random vectors match SGNS.

Validation results below select the better learning rate within each variant. P/R/F denotes macro precision/recall/F1; parameter counts are in millions and include embedding rows. Exact counts remain in the notebook and CSV.

Initialization	Variant	Acc	P/R/F	Total/trainable M	Seconds
tfidf	logistic	0.921	0.921/0.921/0.921	0.400/0.400	39.6
random	trainable	0.920	0.920/0.920/0.920	19.639/19.639	24.1
sgns	frozen	0.895	0.896/0.895/0.895	24.065/0.039	14.6
sgns	finetuned	0.916	0.917/0.916/0.916	24.065/24.065	22.5
gensim	frozen	0.892	0.893/0.892/0.892	24.065/0.039	15.2
gensim	finetuned	0.916	0.917/0.916/0.916	24.065/24.065	23.5
glove	frozen	0.903	0.904/0.903/0.903	40.014/0.013	14.4
glove	finetuned	0.920	0.921/0.920/0.920	40.014/40.014	69.9

The best initialization-specific configuration is selected on validation before official test. Every selected fraction/configuration receives one cached test evaluation. No test score changes model settings.

Initialization	100% variant	Test accuracy	Test F1, 100%	Test F1, 1%
tfidf	logistic	0.923	0.923	0.821
random	trainable	0.922	0.922	0.831
sgns	finetuned	0.919	0.919	0.858
gensim	finetuned	0.918	0.918	0.860
glove	finetuned	0.921	0.921	0.864

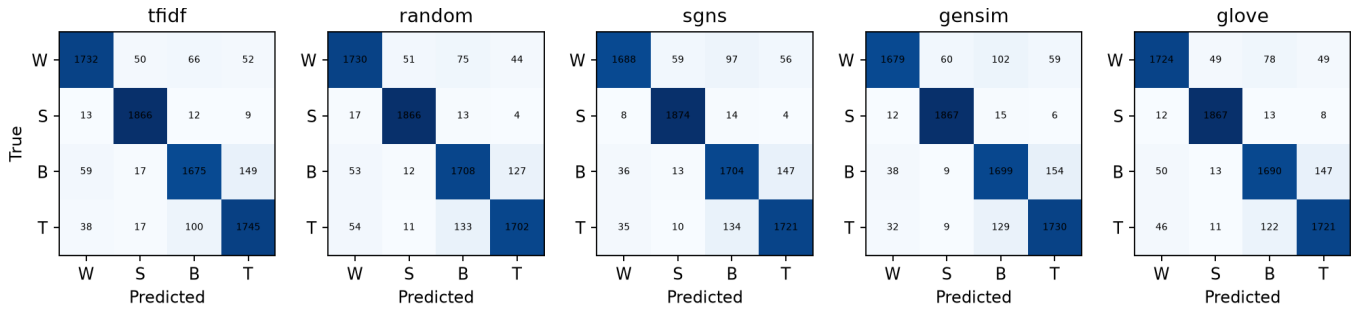


Figure 3: Official test confusion matrices. W/S/B/T: world, sports, business, science/technology.

Variant selections: sgns: finetuned at 100%, finetuned at 1%; gensim: finetuned at 100%, finetuned at 1%; glove: finetuned at 100%, finetuned at 1%. The highest full-data test F1 belongs to tfidf at 0.923. The 1% experiment contains 1,080 training documents, with validation unchanged. Fine-tuning wins validation selection at all four fractions; freezing reduces trainable parameters but never gives the best F1 here. Pretrained vectors do not beat TF-IDF or random vectors at full data, but all three improve over both at 1%. The narrow full-data margins do not establish statistical superiority.

5. Comparison, data efficiency and conclusions

Model	Tokens	Vocabulary	Dim	Train sec	WordSim	SimLex
sgns	40,000,011	80,087	300	450.1	0.680	0.330
gensim	40,000,011	80,087	300	209.9	0.680	0.337
glove	6,000,000,000	400,000	100	NR	0.533	0.298

Model	Add S/T/all	Mul S/T/all	Parallelism	AG OOV %	Test F1
sgns	0.388/0.507/0.469	0.380/0.502/0.464	0.305	3.661	0.919
gensim	0.378/0.512/0.470	0.376/0.510/0.468	0.297	3.661	0.918
glove	0.593/0.679/0.652	0.564/0.664/0.633	0.591	1.044	0.921

S/T denotes semantic/syntactic. SGNS uses the RTX 4050 Laptop 6 GB; gensim uses four CPU workers. The authors report the GloVe corpus as 6B tokens and 400,000 words. Exact released-100D wall time and CPU model are not specified; NR is preferable to substituting a different-dimensional timing example. Similarity coverage is retained in the accompanying comparison CSV and notebook.

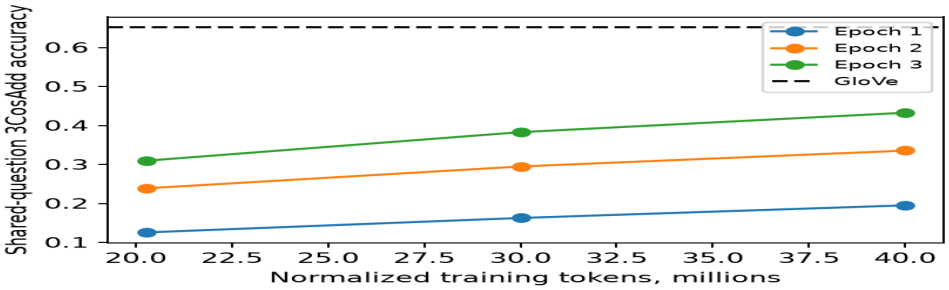


Figure 4: Controlled corpus-size experiment with GloVe reference.

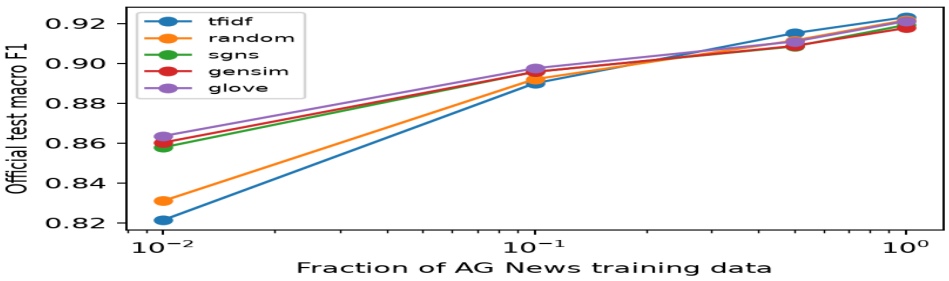


Figure 5: Official test F1 versus labeled training fraction.

SGNS semantic/syntactic accuracy is 0.388/0.507. The overall GloVe minus SGNS gap is +0.183; gensim minus SGNS is +0.001. At epoch three, window 2 versus 5 changes semantic accuracy from 0.339 to 0.247 and syntactic accuracy from 0.475 to 0.465. Ten versus five negatives changes overall accuracy from 0.433 to 0.447.

The corpus-size experiment measures a within-SGNS scaling trend, not a causal decomposition of the GloVe gap. GloVe differs in corpus composition, tokenization, objective and training budget. Small windows reduce semantic accuracy more than syntactic accuracy here, but do not improve absolute syntactic accuracy. More negatives improve overall accuracy under the measured budget, not necessarily every category. Gensim discrepancies remain consistent with the documented optimization differences rather than proof that either objective is implemented incorrectly. Single-seed results, approximate sentence splitting and fixed vocabulary limit generalization.

Sources: Mikolov et al. 2013, arXiv:1310.4546; Pennington et al. 2014, EMNLP, nlp.stanford.edu/pubs/glove.pdf; Stanford original GloVe release; PyTorch Embedding/EmbeddingBag and gensim Word2Vec documentation. Full links and API research are in `research.md` and the notebook. Best original vectors: `informe/artifacts/best_sgns.kv`.